

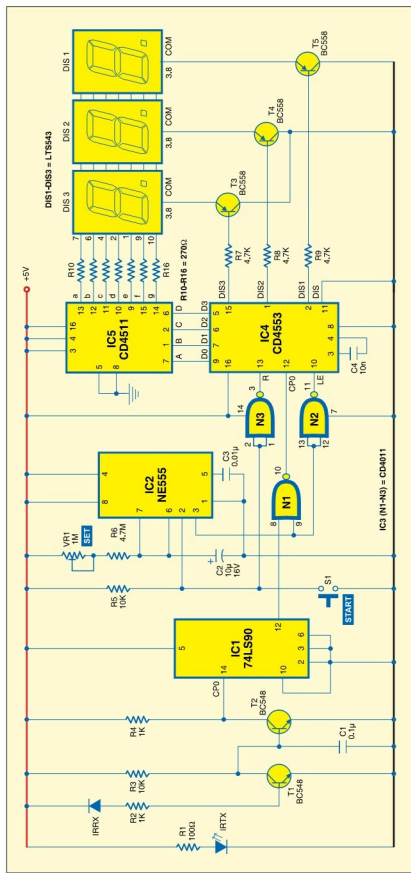


Fig. 2: Circuit of digital respiration rate meter

is used as the clock pulse for 3-digit BCD counter CD4553 (IC4). The output of counter IC1 is connected to input pin 8 of NAND gate N1 of IC3. Input pin 9 of NAND gate N1 is connected to output pin 3 of monostable multivibrator IC2. The time period of the monostable is decided by resistor R6, preset VR1 and capacitor C2. Preset VR1 is used to adjust the time period of the monostable to one minute.

When start switch S1 is pressed, pin 2 of the monostable goes low and it triggers to generate a pulse of 1-minute duration, which is fed to pin 9 of NAND gate N1. Therefore the gate is open for one minute to pass the clock pulses coming from IC1 to pin 12 of the 3-digit counter CD4553. This 1-minute pulse of the monostable is inverted by NAND gate N2, which controls the latch-enable (LE) of IC4553.

When the monostable output goes high, the latch-enable (LE) of CD4553 goes low and the counter advances. When the monostable output goes low, the latch-enable (LE) of IC4 goes high and the counting stops. So there is no further change in the count shown on the 7-segment display.

When start switch S1 is pressed, pin 13 of CD4553 goes high to reset counter CD4553 and the 7-segment display shows '000.' The CD4553 consists of three negative-edge-triggered BCD counters that are cascaded synchronously. A quad latch at the output of each counter permits storage of any given count. The information is then time-division-multiplexed, providing one BCD number or digit at a time. Digit-select outputs provide display control. All the outputs are TTL-compatible. An on-chip oscillator provides low-frequency scanning clock pulse, which drives the multiplexer output selector.

The BCD count outputs of CD4553 (from D0 through D3) are fed to inputs 'A' through 'D' of 7-segment decoder and driver CD4511 (IC5). CD4511 shows the count on 7-segment displays